File 34: GT Intern Workstream — Page 3: AutoCAD Civil 3D Bioengineering Structures Design

> ***Hydrologic Engineering & Geomorphic Design Manual Target School**: Georgia Institute of Technology (Georgia Tech) School of Civil & Environmental Engineering **Project Area**: Upper Savannah Basin & Chattahoochee River Spawning Headwaters **Supervising Board**: Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)*

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1. Design Philosophy: Natural channel Re-Meandering

In high-gradient trout streams, standard concrete retaining walls or rock rip-rap dumping are ecologically unacceptable. They destroy aquatic habitat, block macroinvertebrate colonization, and reflect high velocities downstream, causing accelerated erosion elsewhere.
Instead, you will draft **Natural Channel Design (NCD)** and bioengineered structures using native materials: red cedar logs, oak root-wads, and high-strength geogrid soil lifts. These structures stabilize eroding banks, scour deep cold pools, and mimic historic mountain stream features. Your role as a GT engineering intern is to generate precise, AutoCAD Civil 3D plan-sheets, cross-sections, and detail diagrams.

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2. Technical Drawings & Dimensions for NCD Structures

You will draft four primary bioengineered structures on our site plans:

A. Rock J-Hook Vane (Flow Deflection & Pool Scour)

- **Description**: A rock structure built along the outer bank of a curve. The "vane" portion slopes upstream, directing fast currents away from the bank, while the downstream "hook" scours a deep central pool.
- **AutoCAD Parameters**:
- **Vane Angle**: Set between $20\text{ and } 30\text{ degrees}$ relative to the upstream bank tangent.
- **Slope**: The rock arm must slope down from the bankfull elevation at the bank to the stream bed at $1/3$ the width of the channel.
- **Hook Rocks**: Position the J-hook rocks in the center of the channel, submerged below the baseflow water surface. Leave gaps of $10\text{--}15\text{ cm}$ between rocks to allow trout passage and scour scour halos.

B. Double Log Cross-Vane (Grade Control & Pool Scour)

- **Description**: A two-arm log structure crossing the channel, sloping upstream to create a central plunge pool and prevent bed incision (erosion of the stream bed).
- **AutoCAD Parameters**:

- **Log Material**: Select fresh red cedar or white oak logs ($30\text{--}45\text{ cm}$ diameter).
- **Throat Width**: The central "weir" gap must span exactly $1/3$ the bankfull width of the channel.
- **Log Arm Slope**: Slope the log arms upward at $2\text{--}7\%$ from the stream bed to the bankfull bank line.

C. Cedar Toe-Wood (Bank Stabilization & Woody Habitat)

- **Description**: Layers of overlapping cedar logs and root-wads placed at the toe of a regraded bank curve, covered with soil lifts.
- **AutoCAD Parameters**:
- **Placement**: Place the cedar logs below the baseflow water line to prevent decay from air exposure.
- **Root-wad Projection**: Extend the fibrous cedar root-wads outward into the stream channel at a 45-degree angle facing upstream to trap fine gravels and provide cover for large brown trout.

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3. Structural Anchoring & Geotech Calculations

To ensure these cedar structures do not wash away during major floods, you must execute the **Log Gravity Anchoring Calculations**.

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                      LOG GRAVITY ANCHOR PROFILE

                     BANKFULL WATER STAGE ~~~~~~~~~~~
                                       |
                  [CEDAR LOG]          | Water Depth (H)
                   Volume (V)          |
                                       v
       ============== BEDROCK / CHANNEL BED ==============
                      |                |
        Threaded      |                | Expansion Wedge Bolt
        Epoxy Anchor  |=======#========| (High-Strength Grout)
                              |
                              | Heavy Steel Cable
                              v
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The buoyant upward force (F_b) acting on a submerged log must be completely countered by the downward gravitational force of the soil/rock overburden and mechanical anchor anchors.

Anchor Safety Factor Equation:

$$FS_{anchor} = \frac{W_{overburden} + W_{log\_dry}}{F_{buoyant}} \ge 1.5$$

Where:

- $F_{buoyant} = \gamma_{water} \cdot V_{log}$, where $V_{log} = \pi \cdot R^2 \cdot L$ and $\gamma_{water} = 62.4\text{ lb/ft}^3$.
- $W_{log\_dry} = \gamma_{cedar} \cdot V_{log}$, where dry red cedar weight $\gamma_{cedar} \approx 23.0\text{ lb/ft}^3$.
- $W_{overburden} = \text{weight of rocks/soil placed on top of the log keys in the stream bank}$.

Worked Example:

Calculate the buoyant force and required anchoring weight for a cedar log of radius $R = 0.75\text{ feet}$ and length $L = 12.0\text{ feet}$:

1. Calculate Log Volume (V_{log}):

$$V_{log} = \pi \cdot (0.75)^2 \cdot 12 = 21.21\text{ ft}^3$$

2. Calculate Buoyant Force ($F_{buoyant}$):

$$F_{buoyant} = 62.4 \cdot 21.21 = 1,323.5\text{ lbs of upward force}$$

3. Calculate Log Dry Weight (W_{log_dry}):

$$W_{log\_dry} = 23.0 \cdot 21.21 = 487.8\text{ lbs of downward dry weight}$$

4. Calculate Required Overburden/Mechanical Anchor Weight (W_{req}) for $FS = 1.5$:

$$1.5 = \frac{W_{req} + 487.8}{1,323.5} \implies W_{req} = (1.5 \cdot 1,323.5) - 487.8 = 1,497.45\text{ lbs}$$

Your design must specify a minimum of **1,500 lbs of mechanical cable-anchors** (epoxied into bedrock or secured using concrete/rock deadman anchors in the bank) to satisfy the $FS_{anchor} \ge 1.5$ safety margin.

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4. Geotechnically-Reinforced Soil Lift (GRSL) Details

To rebuild vertical, stable banks along outer bends, you will detail **Geotechnically-Reinforced Soil Lifts (GRSL)**:

- **Core Fabric**: Specify high-durability, double-walled **coir fiber geotextile wrap (Type 700 or 900)**.
- **Filling**: Pack the inside of the fabric wraps with bank-run compacted gravel (15-cm base) and rich organic soil (15-cm top) to promote native root growth.
- **Compaction**: Specify 95% Standard Proctor Density compaction for the soil core.
- **Interlocking**: Wrap the coir fabric around the soil layers in overlapping envelopes, stepping the lifts back at a $1:0.5$ (Horizontal to Vertical) slope. Interleave native willow stakes and tag alder branches between the lifts to bind the system mechanically.